

Statement of Interest — Nauval Zulfikar — PhD Position in Test-Time Adaptation and Agentic AI

The prospect of exploring test-time adaptation and agentic decision-making within the realm of machine learning systems at Aarhus University captivates me. The work of Dr. Behzad Bozorgtabar, particularly his research on adaptive AI systems, aligns closely with my academic pursuits and professional experiences. His paper on "Adaptive Learning in Dynamic Environments" [TODO: verify on <https://scholar.google.com/citations?q=Behzad+Bozorgtabar>] has been particularly inspiring, as it delves into the challenges and methodologies for developing AI systems that can adapt to new data distributions in real-time, a theme that resonates with my interest in creating robust and flexible AI models.

During my MSc in Business Analytics at Aston University, I engaged in several projects that are directly relevant to the themes of this PhD position. My dissertation, "Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis," involved fine-tuning the DeBERTa-v3 transformer on multi-tier supplier review data. This project not only honed my skills in transformer fine-tuning and behavioural text analysis but also provided me with a deep understanding of integrating AI models into larger decision systems. Additionally, my work on the "LLM-Generated Adaptive Shipper Decision Rules" project, where I developed an end-to-end RAG system, further solidified my expertise in creating adaptive AI systems. This project was particularly focused on ensuring reproducibility and ablation, which are crucial for developing reliable AI models that can adapt to changing environments.

I am eager to contribute to Dr. Bozorgtabar's research group by bringing my expertise in machine learning systems and my experience in developing adaptive AI models. I am particularly interested in exploring how test-time adaptation can be leveraged to enhance agentic decision-making in multimodal foundation models. Through this PhD programme, I aim to deepen my understanding of adaptive AI systems and contribute to the development of innovative solutions that can address real-world challenges. I am available to start this position as early as August 2026, aligning with the programme's timeline.